The minimum order of a counterexample to a conjecture on integral generalized sun graphs*

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Abstract

A graph is integral if its adjacency eigenvalues are integers, and a generalized sun graph is a cycle with pendant paths attached to its vertices. Braga, Del-Vecchio and Rodrigues conjectured that the cycle of an integral generalized sun graph that is not a cycle has length divisible by 4. Braga, Moraes and Santos disproved this with the graph $C_{6,1}(0,6,6,12,6,6)$ on 42 vertices, and remarked that their computer search does not show that no smaller counterexample exists. We show that none exists: up to isomorphism, $C_{6,1}(0,6,6,12,6,6)$ is the only counterexample with at most 42 vertices, also when pendant paths of different lengths share a vertex. Every integral generalized sun graph with at most 41 vertices that is not a cycle has a cycle of length 4 and is one of the three graphs found by Braga, Rodrigues and Trevisan in 2017. Up to 42 vertices, spectral lemmas exclude every cycle length $b \not\equiv 0 \pmod 4$ except 3, 6 and 10. The lengths 3 and 6 are settled by two classifications, and the length 10 inside the proof of the main theorem; for these three cases we give no written proof, and they rest on the formal proofs. We also note that if an integral unicyclic graph has a cycle of odd length b, then every prime q with $(q-1) \mid (b-1)$ divides $2b$; hence $3 \mid b$, and $b = 3$ or $b \geq 15$ (after version 1 of this note, Braga, Moraes and Santos proved that only C_3 occurs). Every result of the paper is formally verified in Lean 4. The results were found and formalised by AI coding agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

A graph is *integral* if all eigenvalues of its adjacency matrix are integers, a notion that goes back to Harary and Schwenk [7]. A search of Braga, Rodrigues and Trevisan [4] showed that, besides the cycles C_3, C_4 and C_6, exactly three unicyclic graphs with at most 21 vertices are integral (the three graphs of Theorem 2 below; this classification is also verified in Lean, Section 8). Braga, Del-Vecchio and Rodrigues [1] constructed infinite families of integral unicyclic graphs by attaching pendant paths to the vertices of a cycle; such graphs are called *generalized sun graphs*.

* *Version 2.* Every result of this version is formally verified in Lean 4 (Section 8); version 1 had no formal verification. The theorems are unchanged. Version 1 settled the cycle lengths 3, 6 and 10 by computer searches; version 2 no longer uses them. The cycle lengths 3 and 6 are now settled by two classifications that version 1 obtained only as search results and that are now stated and proved in Lean (Propositions 6.1 and 6.2), and the cycle length 10 inside the formal proof of Theorem 1; the written proofs of Theorems 1 and 2 were rewritten accordingly. Claims of version 1 that have no formal proof have been deleted, among them the counts of search-tree nodes and leaf tests of its Table 1, its tables of program versions and of confirmations, its random tests, and the statement that two programs produced the same set of 52,652 sequences at cycle length 10, none of them integral. The search rules of version 1 are kept only through Lean-proved statements about formally defined sets of sequences (Section 5). The main changes are listed at the end of Section 8.

Following [2], $C_b(n_1P_{t_1},\dots,n_bP_{t_b})$ is obtained from the cycle $v_1\cdots v_bv_1$ by attaching n_k copies of the path P_{t_k} on t_k vertices to v_k (joining v_k to an end of each copy), and $C_{b,t}(n_1,\dots,n_b)$ is the case $t_1=\cdots=t_b=t$. We state the second conjecture of [1] in our words.

Conjecture 2 ([1]; see [2, Conjecture 2])**.** *Let G be an integral generalized sun graph that is not a cycle. Then the cycle of G has length divisible by 4.*

Braga, Moraes and Santos [2] proved the first conjecture of [1] in a strong form: a graph with a pendant path of at least three edges has an eigenvalue in $(1, 2\cos(\pi/9)]$ (and one in $[-2\cos(\pi/9), -1)$), so it is not integral [2, Theorem 2.1]. They disproved Conjecture 2: $C_{6,1}(0,6,6,12,6,6)$ has 42 vertices and spectrum $\{0^{[32]},\pm2^{[2]},\pm3^{[2]},\pm4\}$ [2, Proposition 3.2]. It is the smallest integral member, with $a=6$, of the family $C_{6,1}(0,a,a,2a,a,a)$, $a\ge1$, which contains counterexamples with arbitrarily many vertices [2, Corollary 3.6 and the remark after it]. Their search over boxes of parameter vectors, for graphs with pendant vertices only, found no other integral graph whose cycle length is not divisible by 4. As they point out [2, Section 4], it "does not prove that the graph on 42 vertices has minimum order" among all counterexamples, since vectors outside the boxes may give smaller graphs, and graphs with copies of P_2 were not searched. This note answers that question.

A *counterexample* is a generalized sun graph that is integral, is not a cycle, and has cycle length $b\not\equiv0\pmod4$. We allow any finite number of pendant paths at each cycle vertex, possibly of different lengths at the same vertex. By [2, Theorem 2.1], an integral generalized sun graph then carries only copies of P_1 and P_2. The graphs $C_b(n_1P_{t_1},\dots,n_bP_{t_b})$ form a subclass containing every graph named below, so our results hold in either reading.

Theorem 1. *Up to isomorphism, $C_{6,1}(0,6,6,12,6,6)$ is the only counterexample to Conjecture 2 with at most 42 vertices. In particular, the minimum order of a counterexample is 42.*

Theorem 2. *Every integral generalized sun graph with at most 41 vertices that is not a cycle has a cycle of length 4, and is isomorphic to $C_{4,1}(6,0,3,0)$, with 13 vertices, or to $C_4(5P_2,P_1,2P_2,P_1)$ or $C_{4,2}(4,4,0,0)$, with 20 vertices each.*

These three graphs are the three integral unicyclic graphs found by Braga, Rodrigues and Trevisan [4, Section 5, p. 296, and Table 1, p. 297], with spectra $\{0^{[9]},\pm2,\pm3\}$ and, for the two cospectral graphs, $\{0^{[2]},\pm1^{[7]},\pm2,\pm3\}$; they are pairwise non-isomorphic. Theorem 2 adds that there is no other integral generalized sun graph up to 41 vertices that is not a cycle. The one-page abstract of Veras [13] reports a computer search for integral unicyclic graphs. It names $C_{4,1}(0,16,32,16)$, which has 68 vertices and is integral, and the family $C_{4,1}(0,a,2a,a)$, whose integral members with $a\ge1$ have at least 68 vertices [2]. It does not state the range of the search and names no graph with at most 41 vertices, so nothing it reports overlaps Theorem 2, although the search may have covered part of that range. Write $\mathrm{mult}(\lambda)$ for the multiplicity of an eigenvalue λ. The proofs use the following observation on integral graphs.

Proposition 3. *Let G be an integral graph with e edges that contains an odd cycle, let g be the length of its shortest odd cycles and c_g their number. Then every prime q with $(q-1)\mid(g-1)$ divides $2g\,c_g$; in particular $3\mid g\,c_g$. Moreover, at least $(g-1)/2$ integers $v\ge2$ satisfy $\mathrm{mult}(v)\ne\mathrm{mult}(-v)$, so $2e\ge\sum_{v=2}^{(g+1)/2}v^2$.*

For an integral unicyclic graph with a cycle of odd length b this gives $3\mid b$, and $b=3$ or $b\ge15$, with at least 102 vertices in the latter case (Corollary 3.1).

Method. Up to 42 vertices, hand proofs exclude odd $b\ge5$ (Corollary 3.1), $b\in\{14,18\}$ (Corollary 4.3) and even $b\ge22$ (Lemma 4.1), and a bound due to the first referee agent of version 1 (see the disclosure; Lemma 4.4) excludes every $b\ge14$ again. The cycle length 3 is excluded by Proposition 6.1, the cycle length 6 is settled by Proposition 6.2, and the cycle

length 10 is excluded inside the formal proof of Theorem 1 (Section 6). For these three cases we give no written proof. Version 1 of this note settled them by exhaustive computer searches with separately written programs; version 2 does not use these searches, and every result stated here is proved in Lean 4 (Section 8). Section 5 keeps the mathematics behind the search rules of version 1, as Lean-proved statements about formally defined sets of sequences.

Prior work. The question was raised in version 1 of [2], submitted to the arXiv on 23 September 2026. For version 1 of this note, searches on 26 September 2026 (the arXiv listing of [2], Crossref, MathDB, GitHub, MathOverflow, Mathematics Stack Exchange and web search; not Google Scholar, zbMATH or MathSciNet) found no earlier answer to it and no discussion of it. After version 1 of this note was made public (release v1.6.0 of [6], 27 September 2026), Braga, Moraes and Santos [3] (arXiv:2610.22773v1, submitted 16 October 2026) proved that C_3 is the only integral unicyclic graph whose cycle has odd length, and that $(g-1)!$ divides $2c_g$ for every integral graph of odd girth g with c_g shortest odd cycles. Their first result supersedes Corollary 3.1, and their second implies the first part of Proposition 3 (a prime q with $(q-1) \mid (g-1)$ satisfies $q \le g$). The other parts of Proposition 3, on the eigenvalues $v \ge 2$ with $\mathrm{mult}(v) \ne \mathrm{mult}(-v)$, do not appear there. That paper does not address the minimum order of a counterexample to the conjecture or the integral generalized sun graphs with at most 41 vertices, so Theorems 1 and 2 are not affected. Our searches on 4 November 2026, 12:00–12:17 UTC (the arXiv API and listings since 26 September 2026, Semantic Scholar, OpenAlex, Crossref, DataCite, zbMATH Open, MathDB, MathOverflow and Mathematics Stack Exchange, web search, and public GitHub repositories of AI-assisted mathematics, including `trureturing`, `openai/math` and `formal-conjectures`) found no other work on these questions; the arXiv still lists only version 1 of [2], and Semantic Scholar lists [3] as the only paper citing it. The earlier computer searches for integral unicyclic graphs of [4] and [13] are discussed after Theorem 2. This reports the coverage of our searches, not a guarantee of priority.

2 Notation and spectral lemmas

Throughout, G is a generalized sun graph with cycle $v_0v_1\cdots v_{b-1}v_0$ ($b \ge 3$, indices modulo b) carrying only copies of P_1 and P_2: at v_k hang p_k pendant vertices and q_k pendant paths v_kww'. Put $P = \sum p_k$, $Q = \sum q_k$, $K_p = \{k : p_k > 0\}$, $K_q = \{k : q_k > 0\}$, $Z = \mathbb{Z}_b \setminus K_q$ and $A = A(G)$. Then G has $n = b + P + 2Q$ vertices and n edges. Let $\Theta(G) = \sum_v d_v(d_v-1)$ over all vertices, with degrees d_v; then $\Theta(G) = \sum_k((2+p_k+q_k)(1+p_k+q_k)+2q_k)$. An isomorphism maps the unique cycle onto the unique cycle, and (p_k, q_k) is determined by the tree attached at v_k, so isomorphism classes correspond to dihedral orbits of the sequences $((p_k,q_k))_{k\in\mathbb{Z}_b}$.

For a real symmetric matrix M, $n_+(M)$, $n_-(M)$ and $n_0(M)$ count its positive, negative and zero eigenvalues. For a graph H, $N_{>t}(H)$, $N_{\ge t}(H)$, $N_{<t}(H)$ and $N_{\le t}(H)$ count the eigenvalues of $A(H)$ in the indicated ranges, with multiplicity. If G is integral, B^+ is the multiset of its eigenvalues $\lambda \ge 2$ and $S_2 = \sum_{\lambda\in B^+}\lambda^2$. For nonempty $K = \{k_1 < \cdots < k_t\} \subseteq \{0,\dots,b-1\}$ put $k_{t+1} = k_1 + b$; the *gaps* of K are the pairs (k_i, k_{i+1}), of *length* $k_{i+1} - k_i$. If $|K| = 1$, the only gap has length b.

Lemma 2.1. *For $\lambda \in \mathbb{R} \setminus \{-1, 0, 1\}$ let $f_k(\lambda) = p_k/\lambda + q_k\lambda/(\lambda^2-1)$ and $S(\lambda) = A(C_b) - \lambda I + \mathrm{diag}(f_k(\lambda))_k$.*

(a) If $\lambda > 1$, then $N_{>\lambda}(G) = n_+(S(\lambda))$ and $N_{\ge\lambda}(G) = n_+(S(\lambda)) + n_0(S(\lambda))$.

(b) If $\lambda < -1$, then $N_{<\lambda}(G) = n_-(S(\lambda))$ and $N_{\le\lambda}(G) = n_-(S(\lambda)) + n_0(S(\lambda))$.

(c) $N_{>1}(G) \ge |K_q| + n_+(R)$ and $N_{<-1}(G) \ge |K_q| + n_-(R')$, where R and R' are the principal submatrices of $A(C_b) - I + \mathrm{diag}(p_k)$ and $A(C_b) + I - \mathrm{diag}(p_k)$ on Z.

Proof. With the cycle first, $A-\lambda I = \left(\begin{smallmatrix} A(C_b)-\lambda I & W \\ W^{\mathsf T} & H \end{smallmatrix}\right)$, where $H = H(\lambda)$ is the direct sum of a block $(-\lambda)$ for each pendant vertex and a block $\left(\begin{smallmatrix} -\lambda & 1 \\ 1 & -\lambda \end{smallmatrix}\right)$ for each pendant path (rows w, w'), and W has one entry 1 in the column of each pendant vertex and each w, in the row of its neighbour on the cycle. H has eigenvalues $-\lambda$, $-\lambda \pm 1$, so it is negative (positive) definite for $\lambda > 1$ ($\lambda < -1$). The 2×2 block has inverse with $(1,1)$ entry $-\lambda/(\lambda^2-1)$, so $WH^{-1}W^{\mathsf T} = -\operatorname{diag}(f_k(\lambda))$ and $A - \lambda I$ is congruent to $S(\lambda) \oplus H$; now use Sylvester's law of inertia.

(c) As $\lambda \downarrow 1$, the entries $f_k(\lambda)$, $k \in K_q$, tend to $+\infty$ while the other entries of $S(\lambda)$ stay bounded; so the block S_{KK}, $K = K_q$, is positive definite for λ near 1 and its inverse tends to 0. By Haynsworth's inertia additivity [8], $n_+(S(\lambda)) = |K| + n_+(S_{ZZ} - S_{ZK}S_{KK}^{-1}S_{KZ})$, and the Schur complement tends to R, so its positive inertia is eventually at least $n_+(R)$. Finally $N_{>1}(G) = N_{>\lambda}(G)$ for $\lambda > 1$ close to 1. The case $\lambda \uparrow -1$ is the same, with $f_k(\lambda) \to -\infty$ on K_q. □

Lemma 2.2. *(a) Let $m \in \mathbb{R} \setminus \{-1, 0, 1\}$, $c_k = m - f_k(m)$, $T_k = \left(\begin{smallmatrix} c_k & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $M = T_{b-1} \cdots T_0$. Then $\operatorname{mult}(m) = \dim\ker(M - I) \le 2$; it equals 2 if and only if $M = I$, and 1 if and only if $M \ne I$ and $\operatorname{tr} M = 2$.*

(b) $\operatorname{mult}(0) = P - |K_p| + \delta_0$, where, if $K_p = \emptyset$, $\delta_0 = 2$ when $4 \mid b$ and $\delta_0 = 0$ otherwise, and, if $K_p \ne \emptyset$, δ_0 is the number of gaps of K_p of even length.

(c) For $\varepsilon = \pm 1$, $\operatorname{mult}(\varepsilon) = Q - |K_q| + \delta_\varepsilon$, where δ_ε is the dimension of the space of $y \in \mathbb{R}^{\mathbb{Z}_b}$ with $y_k = 0$ for $k \in K_q$ and $y_{k-1} + y_{k+1} = \varepsilon(1 - p_k)y_k$ for $k \notin K_q$. Each gap of K_q contributes 0 or 1 to δ_ε, and $\delta_\varepsilon \le 2$ if $K_q = \emptyset$.

(d) $\operatorname{mult}(0) \le \max(P, 2)$ and $\operatorname{mult}(\pm 1) \le \max(Q, 2)$.

Proof. (a) If $Ax = mx$, then $x_a = x_{v_k}/m$ at a pendant vertex a of v_k, and $x_{w'} = x_w/m$, $x_w = mx_{v_k}/(m^2-1)$ on a pendant path $v_k w w'$. So x is determined by its restriction y to the cycle, which satisfies $y_{k-1} + y_{k+1} = c_k y_k$, and every solution y extends to an eigenvector. Writing $(y_{k+1}, y_k)^{\mathsf T} = T_k(y_k, y_{k-1})^{\mathsf T}$, the solutions correspond to the vectors fixed by M; as $\det M = 1$, $M - I$ is singular if and only if $\operatorname{tr} M = 2$.

(b) If $Ax = 0$, a pendant vertex at v_k forces $x_{v_k} = 0$, and a pendant path gives $x_w = 0$, $x_{w'} = -x_{v_k}$. For $k \in K_p$ the equation at v_k only fixes the sum of the p_k pendant values ($p_k - 1$ free values), and for $k \notin K_p$ it reads $y_{k-1} + y_{k+1} = 0$. So δ_0 is the dimension of the space of y with $y_k = 0$ on K_p and $y_{k-1} + y_{k+1} = 0$ off K_p. If $K_p = \emptyset$, then $y_{k+2} = -y_k$ for all k, with a 2-dimensional space of b-periodic solutions if $4 \mid b$ and none otherwise. If $K_p \ne \emptyset$, the equations split along the gaps. On a gap of length d put $z_j = y_{k_i+j}$; then $z_0 = z_d = 0$ and $z_{j-1} + z_{j+1} = 0$ for $0 < j < d$, which forces $z_j = 0$ for even j and $z_{j+2} = -z_j$ for odd j. For even d this leaves z_1 free; for odd d it gives $z_1 = \pm z_d = 0$.

(c) If $Ax = \varepsilon x$, a pendant path gives $x_{w'} = \varepsilon x_w$ and $\varepsilon x_w = x_{v_k} + x_{w'}$, so $x_{v_k} = 0$ for $k \in K_q$, and the equation at v_k only fixes the sum of the q_k values x_w ($q_k - 1$ free values). A pendant vertex gives $x_a = \varepsilon x_{v_k}$, so for $k \notin K_q$ the equation at v_k reads $y_{k-1} + y_{k+1} + \varepsilon p_k y_k = \varepsilon y_k$. On a gap of K_q the recurrence determines all values from the first one after the left end, so the gap contributes at most 1; if $K_q = \emptyset$ the periodic solutions of a recurrence of order two form a space of dimension at most 2.

(d) If $K_p \ne \emptyset$, then $\delta_0 \le |K_p|$, the number of gaps, so $\operatorname{mult}(0) \le P$; otherwise $\operatorname{mult}(0) = \delta_0 \le 2$. Likewise for ± 1. □

Lemma 2.3. *Let H be a graph with e edges, degrees d_v, c_4 cycles of length 4, and adjacency matrix A.*

(a) $\operatorname{tr} A^2 = 2e$ and $\operatorname{tr} A^4 = 2e + 2\sum_v d_v(d_v - 1) + 8c_4$.

(b) If the shortest odd cycles of H have length g and there are c_g of them, then $\operatorname{tr} A^L = 0$ for odd $L < g$ and $\operatorname{tr} A^g = 2g\,c_g$.

Proof. We count closed walks. (a) A closed walk of length 4 traverses one edge four times (2 walks per edge), or two edges with a common vertex twice each (4 walks for each of the $\sum_v \binom{d_v}{2}$ such pairs), or a 4-cycle (8 walks per cycle). (b) A closed walk of odd length L that repeats a vertex splits there into two shorter closed walks, one of odd length. By induction it contains an odd cycle of length at most L, with equality only if it traverses a cycle once. So no closed walk of odd length $L < g$ exists, and each g-cycle gives $2g$ closed walks of length g. □

For our graphs $e = n$, $c_4 = [b = 4]$, $\sum_v d_v(d_v - 1) = \Theta(G)$, and $g = b$, $c_g = 1$ if b is odd. If G is integral, the eigenvalues in $\{0, \pm 1\}$ satisfy $\lambda^4 = \lambda^2$ and $\lambda^L = \lambda$ for odd L, and $\operatorname{tr} A = 0$, so

$$\sum_\lambda \lambda^2 = 2n, \qquad \sum_{|\lambda|\geq 2} \lambda^2(\lambda^2 - 1) = 2\,\Theta(G) + 8\,[b = 4], \tag{1}$$

$$\sum_{|\lambda|\geq 2}(\lambda^L - \lambda) = 0 \;\; (L \text{ odd},\; 3 \leq L < b), \qquad \sum_{|\lambda|\geq 2}(\lambda^b - \lambda) = 2b \;\; \text{if } b \text{ is odd}. \tag{2}$$

Lemma 2.4. *Let G be integral and not a cycle.*

(a) The largest eigenvalue ρ is simple and $\rho \geq 3$; every eigenvalue satisfies $|\lambda| \leq \rho$ and $|\lambda| \leq \sqrt{2n}$, and $\lambda > -\rho$ if b is odd.

(b) $\operatorname{mult}(\lambda) \leq 2$ whenever $|\lambda| \geq 2$.

(c) If b is even, the spectrum is symmetric, $S_2 = n - \operatorname{mult}(1)$ and $P + Q \leq S_2 - b + 2$. If b is odd, $P + Q \leq \frac{1}{2}\sum_{|\lambda|\geq 2} \lambda^2 - b + 2$.

(d) If b is even, then $|B^+| \leq 4$ if $n \leq 41$, and $|B^+| \leq 5$ if $n \leq 42$, with equality only if $B^+ = \{4, 3, 3, 2, 2\}$.

Proof. (a) By Perron–Frobenius theory [5], ρ is simple, $|\lambda| \leq \rho$, and $-\rho$ is an eigenvalue only if G is bipartite. As G properly contains C_b, $\rho > 2$. Also $\lambda^2 \leq \operatorname{tr} A^2 = 2n$. (b) is Lemma 2.2(a). (c) For even b, G is bipartite and (1) gives $2n = 2S_2 + 2\operatorname{mult}(1)$; by Lemma 2.2(d), $S_2 \geq b + P + 2Q - \max(Q, 2) \geq b + P + Q - 2$. For odd b, $\sum_{|\lambda|\geq 2} \lambda^2 = 2n - \operatorname{mult}(1) - \operatorname{mult}(-1)$, and the same estimate applies. (d) By (a) and (b), B^+ has a simple largest element and contains every other value at most twice. Five elements need $\rho \geq 4$ and $S_2 \geq 4^2 + 2 \cdot 3^2 + 2 \cdot 2^2 = 42$, with equality only for $\{4, 3, 3, 2, 2\}$, and six need $S_2 \geq 5^2 + 4^2 + 2 \cdot 3^2 + 2 \cdot 2^2 = 67$; and $S_2 \leq n$ by (c). □

3 Odd cycles

Proof of Proposition 3. For $v \geq 2$ let $\eta_v = \operatorname{mult}(v) - \operatorname{mult}(-v)$. For odd L the eigenvalues in $\{0, \pm 1\}$ satisfy $\lambda^L = \lambda$, and $(-v)^L - (-v) = -(v^L - v)$, so by Lemma 2.3(b) and $\operatorname{tr} A = 0$,

$$\sum_{v\geq 2} \eta_v\,(v^L - v) = \operatorname{tr} A^L = 0 \;\; (L \text{ odd},\; L < g), \qquad \sum_{v\geq 2} \eta_v\,(v^g - v) = 2g\,c_g. \tag{3}$$

Let q be prime with $(q-1) \mid (g-1)$. For every integer λ, $\lambda^g \equiv \lambda \pmod q$: this is clear if $q \mid \lambda$, and otherwise $\lambda^{q-1} \equiv 1$ by Fermat's little theorem, so $\lambda^g = \lambda(\lambda^{q-1})^{(g-1)/(q-1)} \equiv \lambda$. Hence q divides $\sum_\lambda(\lambda^g - \lambda) = 2g\,c_g$. The prime $q = 3$ qualifies as g is odd.

Let $V = \{v \geq 2 : \eta_v \neq 0\}$, $s = |V|$, and suppose $s \leq (g-3)/2$. The exponents $L = 2j + 1$, $1 \leq j \leq s$, are odd and less than g, so $\sum_{v\in V} \eta_v v(w_v^j - 1) = 0$ for $1 \leq j \leq s$, with $w_v = v^2$. Subtracting the row of the value 1 from the other rows of the Vandermonde matrix of the distinct numbers 1 and w_v ($v \in V$) shows that $\det(w_v^j - 1)$ equals that Vandermonde determinant, which is nonzero. Hence all $\eta_v = 0$, which contradicts the definition of V if $s > 0$ and (3) if $s = 0$. So $s \geq (g-1)/2$, and $2e = \sum_\lambda \lambda^2 \geq \sum_{v\in V} v^2 \geq \sum_{v=2}^{(g+1)/2} v^2$. □

Corollary 3.1. *Let G be an integral unicyclic graph with n vertices whose cycle has odd length b, and let κ_b be the product of the primes q with $(q-1) \mid (b-1)$. Then $\kappa_b \mid 2b$; in particular $3 \mid b$. The odd $b < 200$ with $\kappa_b \mid 2b$ are 3, 15, 27, 39, 63, 75, 87, 99, 123, 135, 147, 159, 183 and 195. If $b \neq 3$, then $b \geq 15$ and $n \geq 102$.*

After version 1 was released, Braga, Moraes and Santos [3] proved that C_3 is the only integral unicyclic graph with a cycle of odd length, which supersedes Corollary 3.1; we keep it with its proof, which is formalized.

Proof. Here $g = b$, $c_g = 1$ and $e = n$, so $\kappa_b \mid 2b$ by Proposition 3. For $b = 5, 7, 9, 11, 13$, $\kappa_b = 30, 42, 30, 66, 2730$ does not divide $2b$; the list is a direct computation. If $b \geq 15$, then $2n \geq \sum_{v=2}^{8} v^2 = 203$. □

Remark 3.2. By the theorem of von Staudt and Clausen, κ_b is the denominator of the Bernoulli number B_{b-1} for odd $b \geq 3$. The first part of Proposition 3 uses integrality; for an arbitrary integer matrix A and a prime q one has the classical congruence $\operatorname{tr} A^q \equiv \operatorname{tr} A \pmod{q}$. The second part is a variant of the odd-polynomial argument of van Dam and Haemers [12, Section 2]. Version 1 reported that searches on 26 September 2026 found no earlier statement of Proposition 3, even for unicyclic graphs, but that the paper of Omidi [10] on necessary conditions for integral unicyclic graphs and [1] could not be read in full. As the argument is elementary, we present it as an observation.

4 Restricting the cycle length

Lemma 4.1. *$N_{>1}(G) \geq N_{>1}(C_b) = 2\lceil b/6 \rceil - 1$ and $N_{<-1}(G) \geq N_{<-1}(C_b)$. So if G is integral and not a cycle, then $|B^+| \geq 2\lceil b/6 \rceil - 1$; if moreover b is even, then $b \leq 12$ when $n \leq 41$, and $b \leq 18$ when $n \leq 42$, with $B^+ = \{4, 3, 3, 2, 2\}$ for $b \in \{14, 16, 18\}$. In particular, if $n \leq 42$ and $b \equiv 2 \pmod 4$, then $b \in \{6, 10, 14, 18\}$.*

Proof. C_b is an induced subgraph of G, so by Cauchy interlacing [5] the i-th largest (smallest) eigenvalue of G is at least (at most) that of C_b for $i \leq b$. The eigenvalues of C_b are $2\cos(2\pi j/b)$, $j \in \mathbb{Z}_b$, and $2\cos(2\pi j/b) > 1$ if and only if $6|j| < b$ for $j \in (-b/2, b/2]$, which holds for $2\lceil b/6 \rceil - 1$ values of j. If G is integral, its eigenvalues greater than 1 form B^+. Now apply Lemma 2.4(d); $2\lceil b/6 \rceil - 1 = 5$ for $13 \leq b \leq 18$. □

Lemma 4.2. *Let G be integral with b even, and $s = |B^+|$. Then $\operatorname{mult}(\pm 1) = n - S_2$ and $\operatorname{mult}(0) = 2S_2 - n - 2s$. Hence $S_2 \leq n \leq 2S_2 - 2s$ and $2S_2 - n - 2s \leq \max(P, 2)$.*

Proof. By symmetry $n = \operatorname{mult}(0) + 2\operatorname{mult}(1) + 2s$, and (1) gives $2n = 2S_2 + 2\operatorname{mult}(1)$. The last inequality is Lemma 2.2(d). □

Corollary 4.3. *No integral generalized sun graph with cycle length 14 or 18 has at most 42 vertices.*

Proof. The cycles C_{14} and C_{18} are not integral, as $2\cos(2\pi/b)$ is not an integer for these b. Otherwise, by [2, Theorem 2.1] the graph carries only copies of P_1 and P_2, so it is a graph G as in Section 2 (see the remark before the proof of Theorem 1). By Lemma 4.1, $s = 5$ and $S_2 = 42$. By Lemma 4.2, $42 \leq n \leq 42$, so $\operatorname{mult}(1) = 0$ and $\operatorname{mult}(0) = 84 - 42 - 10 = 32 \leq \max(P, 2)$. Then $n \geq b + P \geq 46$, a contradiction. □

The next bound is due to the first referee agent of version 1 (see the disclosure); it was the pruning rule of implementation C of version 1 (Section 5). For $t \geq 1$ let $\sigma(t)$ be the sum of the first t terms of $4, 4, 4, 4, 9, 9, 9, 9, 16, \ldots$, in which each v^2, $v \geq 2$, appears four times, and let $\sigma(t) = 0$ for $t \leq 0$. Thus $\sigma(9) = 68$, $\sigma(10) = 84$, $\sigma(11) = 100$, $\sigma(14) = 166$.

Lemma 4.4. *Let G be integral and not a cycle, r the number of eigenvalues with $|\lambda| \geq 2$, and $r_0 = b + |K_p| - \delta_0$, with $\delta_0, \delta_{\pm 1}$ as in Lemma 2.2. Then*

(a) $r = r_0 + 2|K_q| - \delta_1 - \delta_{-1} \geq b - 4$;

(b) $\sigma(r) \leq 2n$;

(c) $\sigma(r_0) \leq 2n - 2|K_q|$ *if* $K_q \neq \emptyset$, *and* $\sigma(r_0 - 4) \leq 2n - 4$ *if* $K_q = \emptyset$.

Consequently, $b \leq 13$ *if* $n \leq 42$.

Proof. (a) Since $r = n - \mathrm{mult}(0) - \mathrm{mult}(1) - \mathrm{mult}(-1)$, the formula follows from Lemma 2.2(b),(c). If $K_p \neq \emptyset$, then $\delta_0 \leq |K_p|$ and $r_0 \geq b$; otherwise $K_q \neq \emptyset$ and $r_0 \geq b - 2$. With $\delta_{\pm 1} \leq |K_q|$ if $K_q \neq \emptyset$ and $\delta_{\pm 1} \leq 2$ otherwise, $r \geq b - 4$. (b) Each value $\pm v$, $v \geq 2$, occurs at most twice (Lemma 2.4(b)), so the squares of these r eigenvalues sum to at least $\sigma(r)$ and at most $2n$. (c) If $K_q \neq \emptyset$, put $x = 2|K_q| - \delta_1 - \delta_{-1} \geq 0$; then $r = r_0 + x$ and $\mathrm{mult}(1) + \mathrm{mult}(-1) = 2Q - x$, so $\sigma(r_0) + 4x \leq \sigma(r) \leq 2n - 2Q + x$, whence $\sigma(r_0) \leq 2n - 2|K_q|$. If $K_q = \emptyset$, put $y = \delta_1 + \delta_{-1} \leq 4$; then $r = r_0 - y$ and $\sigma(r_0 - 4) + 4(4 - y) \leq \sigma(r_0 - y) \leq 2n - y$ (trivially if $r_0 < 4$), whence $\sigma(r_0 - 4) \leq 2n - 4$.

Let $n \leq 42$. If $b \geq 15$, (a) and (b) give $100 = \sigma(11) \leq 84$. If $b = 14$, then $r_0 \geq 14$ (as $4 \nmid 14$), and (c) gives $166 \leq 82$ or $84 \leq 80$. □

5 The search rules of version 1

Version 1 settled the cycle lengths 3, 6 and 10, and confirmed the other cases, by exhaustive computer searches with four separately written programs, called A, B (with a variant B2), C and D there; C and D were written by referee agents. As described in version 1, each program ran through the sequences $((p_k, q_k))_{k \in \mathbb{Z}_b}$ with $n \leq N$, discarded sequences by spectral necessary conditions, and tested the remaining graphs exactly for integrality. Version 2 does not use these searches, and we make no claim about the programs or their runs; the programs, logs and referee reports remain in the directory `certificates/sun-graphs` of version 1.6.0 of [6]. This section keeps the mathematics behind their rules: the bounds used for discarding partial assignments (Section 5.1), the exact tests for integrality (Section 5.2), and properties of the sets of sequences that the rules define, stated for formal definitions of these sets (Section 5.3). Everything stated in this section is proved in Lean.

5.1 Soundness of the pruning

Let $X = \{0, \ldots, b - 2\}$. Below, a tridiagonal matrix indexed by $Y \subseteq X$ has off-diagonal entry 1 in the positions $(k, k+1)$ and $(k+1, k)$ for which both k and $k+1$ lie in Y, and 0 elsewhere.

Lemma 5.1. *Let $m \geq 2$ and let T be tridiagonal on X with diagonal entries $t_k \leq f_k(m) - m$. Then $N_{>m}(G), N_{<-m}(G) \geq n_+(T)$ and $N_{\geq m}(G), N_{\leq -m}(G) \geq n_+(T) + n_0(T)$. If $Y \subseteq K_q \cap X$ and T' is tridiagonal on $X \setminus Y$ with diagonal entries $t'_k \leq p_k - 1$, then $N_{>1}(G), N_{<-1}(G) \geq |Y| + n_+(T')$.*

Proof. The principal submatrix $S(m)_X$ is tridiagonal, as $b - 1 \notin X$, and $S(m)_X - T$ is diagonal and nonnegative. By Weyl's inequalities and Cauchy interlacing [5], $n_+(S(m)) \geq n_+(S(m)_X) \geq n_+(T)$, and likewise for $n_+ + n_0$; apply Lemma 2.1(a). As f_k is odd, $\Delta S(-m)_X \Delta = -S(m)_X$ for $\Delta = \mathrm{diag}((-1)^k)$, and Lemma 2.1(b) gives the negative side. For T', let $U = (K_q \cap X) \setminus Y$. By Lemma 2.1(c) and interlacing, $N_{>1}(G) \geq |K_q| + n_+(R_{Z \cap X})$; and $R_{Z \cap X}$ arises from the tridiagonal matrix on $X \setminus Y$ with diagonal $p_k - 1$ by deleting the indices in U, so $n_+(R_{Z \cap X}) \geq n_+(T') - |U|$, while $|K_q| \geq |Y| + |U|$. The negative side is the same with R' and Δ. □

The node test of A (in its formal model, Section 5.3) applies these bounds to partial assignments. With $v_0, \dots, v_{d-1}$ assigned ($d \le b-1$), T and T' use the assigned values and, elsewhere, those of a bare vertex ($t_k = -m$, $t'_k = -1$), with $Y = K_q \cap \{0, \dots, d-1\}$. Raising p_k or q_k raises t_k, t'_k or moves k into Y, which does not decrease $|Y| + n_+(T')$; so the bounds are monotone (Proposition 5.7(d)). According to version 1, A computed the inertias by the pivot recurrence $d_1 = t_1$, $d_k = t_k - 1/d_{k-1}$, restarted after each index missing from the index set (a tridiagonal matrix on an index set with gaps is block diagonal); the next lemma shows that this recurrence gives the inertia.

Lemma 5.2. *Let T be symmetric tridiagonal with nonzero off-diagonal entries. If an exact zero pivot $d_k = 0$ is counted as negative, d_{k+1} as positive, and the recurrence is restarted with $d_{k+2} = t_{k+2}$, then the number of positive pivots is $n_+(T)$; with the signs exchanged one obtains $n_-(T)$.*

Proof. For small $\varepsilon > 0$, $n_+(T) = n_+(T - \varepsilon I)$, whose pivots are $\Delta_k(\varepsilon)/\Delta_{k-1}(\varepsilon)$ with leading minors Δ_k. The leading blocks are Jacobi matrices, with simple and strictly interlacing eigenvalues; if $\Delta_k(0) = 0$, counting their negative eigenvalues shows that $\Delta_k(\varepsilon)$ and $\Delta_{k-1}(0)$ have opposite signs. The recurrence $\Delta_{j+1} = t_{j+1}\Delta_j - a_j^2\Delta_{j-1}$ ($a_j \ne 0$ the off-diagonal entries) gives $\Delta_{k+1}(0) = -a_k^2\Delta_{k-1}(0)$ and $\Delta_{k+2}(0) = t_{k+2}\Delta_{k+1}(0)$. So $d_k(\varepsilon) \to 0^-$, $d_{k+1}(\varepsilon) \to +\infty$, $d_{k+2}(\varepsilon) \to t_{k+2}$. For n_- use $T + \varepsilon I$. □

Lemma 5.3. *If $v_0, \dots, v_{d-1}$ are assigned, contributing Θ_d to $\Theta(G)$ and W to $n-b$, and $r = b-d$, then every completion with $n \le N$ has $\Theta_d + 2r \le \Theta(G) \le \Theta_d + g(N-b-W) + 2(r-1)$, where $g(w) = (2+w)(1+w)$.*

Proof. A vertex of weight $w = p+2q$ contributes $g(w-q)+2q \le g(w)$, as $g(t)-g(t-1) = 2t+2$, and at least $g(0) = 2$. As g is convex, $g(a) + g(c) \le g(a+c) + g(0)$, and g is increasing. □

A partial assignment is discarded by this test if no admissible value of $\Theta(G)$ from (1) lies in the window. The formal model of version 6 of A also uses the following consequence of Lemma 4.2.

Lemma 5.4. *Let b be even, G integral with $n \le N$, $s = |B^+|$ and $n_{\max} = \min(N, 2S_2 - 2s)$. Let $d \le b$, and let P_d and Q_d be the sums of p_k and q_k over $k < d$. If $P_d \ge 2$, then $2Q_d \le 2n_{\max} - b - 2S_2 + 2s$.*

Proof. By Lemma 4.2, $n \le 2S_2 - 2s$, so $n \le n_{\max}$, and $2S_2 - n - 2s \le \max(P, 2) = P$, as $P \ge P_d \ge 2$. Hence $n = b + P + 2Q \ge b + 2S_2 - n - 2s + 2Q$, that is, $2Q \le 2n - b - 2S_2 + 2s$, and $Q_d \le Q$. □

5.2 Exact tests

Version 1 describes exact tests for integrality at the leaves of the searches; they rest on the following facts. For a prime ℓ and a finite set $S \subset \mathbb{Z}$ whose elements have distinct residues modulo ℓ, and a monic $f \in \mathbb{Z}[x]$, let $\mu_\ell(S, f)$ be the sum over $s \in S$ of the multiplicity of the root s of f modulo ℓ. Put

$$F(x) = \det(x(x^2-1)(xI - A(C_b)) - \operatorname{diag}(p_k(x^2-1) + q_k x^2)).$$

For an integer m let $\beta_\ell(m)$ be the nullity modulo ℓ of the following integer $b \times b$ matrix, plus an offset: for $m \notin \{0, \pm 1\}$ the matrix $m(m^2-1)S(m) = m(m^2-1)(A(C_b) - mI) + \operatorname{diag}(p_k(m^2-1) + q_k m^2)$ with no offset; for $m = 0$ the matrix of the system of Lemma 2.2(b) ($y_k = 0$ for $k \in K_p$, $y_{k-1} + y_{k+1} = 0$ otherwise) with the offset $P - |K_p|$; for $m = \varepsilon = \pm 1$ that of the system of Lemma 2.2(c) with the offset $Q - |K_q|$.

Proposition 5.5. *(a) For every finite graph H, $A(H)$ has $|V(H)|$ real eigenvalues counted with multiplicity;* $\mathrm{mult}(m)$ *is the nullity over $\mathbb{Q}$ of $A(H) - mI$ for every integer m; H is integral if and only if* $\det(xI - A(H))$ *is a product of factors $x - z$, $z \in \mathbb{Z}$; and if H is integral with all eigenvalues in a finite set $S \subset \mathbb{Z}$, then $\sum_{m\in S} \mathrm{mult}(m) = |V(H)|$.*

(b) If $n \le 49$, every eigenvalue λ of G satisfies $\lambda^2 \le 2n$ and $|\lambda| < 10$. If moreover G is integral, then $\det(xI - A)$ *and F are products of factors $x - z$ with $z \in [-9, 9]$, and $\sum_{m=-9}^{9} \mathrm{mult}(m) = n$.*

(c) $x^P(x^2-1)^Q F(x) = x^b(x^2-1)^b \det(xI - A)$, and F is monic of degree $4b$.

(d) $\mathrm{mult}(m)$ *is the nullity over $\mathbb{Q}$ of $m(m^2-1)S(m)$ for every integer $m \notin \{0, \pm 1\}$, and* $\mathrm{mult}(0)$ *and* $\mathrm{mult}(\pm 1)$ *are the nullities over $\mathbb{Q}$ of the systems of Lemma 2.2(b),(c) plus the offsets $P - |K_p|$ and $Q - |K_q|$. For every integer square matrix and every prime ℓ, the rank modulo ℓ is at most the rank over $\mathbb{Q}$, and ranks and nullities over $\mathbb{Q}$ and over $\mathbb{R}$ agree. Hence* $\mathrm{mult}(m) \le \beta_\ell(m)$ *for every prime ℓ and every integer m.*

(e) Let ℓ be prime, $f \in \mathbb{Z}[x]$ monic and $S \subset \mathbb{Z}$ finite with distinct residues modulo ℓ. Then f modulo ℓ is monic of the same degree, the multiplicity of a root $z \in \mathbb{Z}$ of f does not drop modulo ℓ, and if f is a product of factors $x - s$ with $s \in S$, then $\mu_\ell(S, f) = \deg f$. If $\ell > 2R$, the integers in $[-R, R]$ have distinct residues modulo ℓ.

(f) The numbers 10^9+7, 998244353, $2^{61}-1$, $10^{18}+3$ and $2^{31}-1$ are primes greater than 216. Let $n \le 49$ and let G be integral. Then $\mu_\ell([-9, 9], F) = 4b$ for $\ell \in \{10^9+7, 998244353, 2^{61}-1\}$, $\mu_\ell([-10, 10], \det(xI - A)) = n$ for $\ell \in \{2^{61} - 1, 10^{18} + 3\}$, and $n \le \sum_{m=-9}^{9} \beta_\ell(m)$ for $\ell = 2^{31} - 1$. So if one of these equalities or this inequality fails, G is not integral.

(g) For a prime ℓ and $d < \ell$, the nodes $20, 21, \ldots, 20 + d$ are distinct modulo ℓ, and Newton's divided differences on these nodes recover every polynomial of degree at most d over $\mathbb{Z}/\ell$ from its values. For $\ell \in \{10^9 + 7, 998244353, 2^{61} - 1\}$ and $3 \le b \le 49$, the determinants of the matrix defining F, evaluated modulo ℓ at these nodes, are the values of F modulo ℓ, and the reconstruction of degree $4b$ is F modulo ℓ.

(h) For every finite unicyclic graph H and every edge $e = uv$ on its cycle C, $\det(xI - A(H)) = \phi(H - e) - \phi(H - u - v) - 2\phi(H - V(C))$, where $\phi(K) = \det(xI - A(K))$ [11]. For G and $e = v_0 v_{b-1}$, $\phi(G - V(C_b)) = \prod_k x^{p_k}(x^2-1)^{q_k}$, and $\phi(G - e)$ and $\phi(G - v_0 - v_{b-1})$ are given by an explicit two-term recurrence over the pairs (p_k, q_k).

The identity (c) follows from the proof of Lemma 2.1. According to version 1, B computed F modulo the three primes of (f) by interpolation, as in (g), C computed $\det(xI - A)$ modulo its two primes by the formula (h), and D computed the bounds $\beta_\ell(m)$ for $\ell = 2^{31} - 1$; each discarded a graph when the corresponding test of (f) failed. Passing these tests does not imply integrality, and nothing here says that it does.

According to version 1, implementation A decided integrality with Lemma 2.2(a), in integer arithmetic modulo four primes. For an integer m with $|m| \ge 2$ put $h = m(m^2-1)$, $S_k = hc_k = m^2(m^2-1) - p_k(m^2-1) - q_k m^2$ and $\tilde{T}_k = hT_k = \left(\begin{smallmatrix} S_k & -h \\ h & 0 \end{smallmatrix}\right)$, so that $\tilde{M} = \tilde{T}_{b-1} \cdots \tilde{T}_0 = h^b M$, and let $E = \prod_k (|S_k| + |h|)$. By Lemma 2.2(a), $\mathrm{mult}(m) = 2$ if $\tilde{M} = h^b I$, $\mathrm{mult}(m) = 1$ if $\tilde{M} \ne h^b I$ and $\operatorname{tr} \tilde{M} = 2h^b$, and $\mathrm{mult}(m) = 0$ otherwise.

Proposition 5.6. *(a) The numbers $\ell_1 = 2^{62} - 57$, $\ell_2 = 2^{62} - 87$, $\ell_3 = 2^{62} - 117$ and $\ell_4 = 2^{62} - 143$ are pairwise distinct primes, and $\ell_1\ell_2\ell_3\ell_4 > 2^{240}$. An integer z with $|z| < \ell_1\ell_2\ell_3\ell_4$ is 0 if and only if $\ell_i \mid z$ for $i = 1, \ldots, 4$.*

(b) For all b, p, q and $|m| \ge 2$, $\tilde{M}$ is h^b times the real matrix M of Lemma 2.2(a), every entry of $\tilde{M}$ has absolute value at most E, $|\operatorname{tr} \tilde{M} - 2h^b| \le 2E + 2|h|^b$, and every entry of $\tilde{M} - h^b I$ has absolute value at most $E + |h|^b$.

(c) For each of the 23 *runs of A reported in version 1 (a version of A and a pair* (b, N)*), every sequence in the formal leaf set of that version at* (b, N) *(Section 5.3) and every* m *with* $2 \le |m| \le 9$*, we have* $2E + 2|h|^b < 2^{240}$*. Hence* $\operatorname{tr} \tilde{M} = 2h^b$ *if and only if* $\operatorname{tr} \tilde{M} \equiv 2h^b \pmod{\ell_i}$ *for all* i*,* $\tilde{M} = h^b I$ *if and only if* $\tilde{M} \equiv h^b I \pmod{\ell_i}$ *for all* i*, and the multiplicity computed from these residues as above equals* $\operatorname{mult}(m)$*.*

5.3 Formal search sets

The following definitions model the rules of the programs of version 1. They are the definitions of the Lean statement files, which state them in full; we make no claim that the programs implement them. A *sequence* for (b, N) is a pair (p, q) of maps $\mathbb{Z}_b \to \mathbb{N}$ with $P + 2Q > 0$ and $n = b + P + 2Q \le N$, so that $G = G(p, q)$ is a graph of Section 2 that is not a cycle.

Definition. A *feature vector* is a pair $s = (s^+, s^-)$ of maps $\{2, \dots, 9\} \to \{0, 1, 2\}$, standing for $v \mapsto \operatorname{mult}(v)$ and $v \mapsto \operatorname{mult}(-v)$. Put $\epsilon_b = 0$ for even b and $\epsilon_b = 1$ for odd b, $s_v = s^+(v) + \epsilon_b s^-(v)$, $S_2(s) = \sum_v s_v v^2$ and $M_4(s) = \sum_v s_v v^2(v^2 - 1)$. The vector s is *admissible* for (b, N) if some $\rho \ge 3$ has $s^+(\rho) = 1$ and $s^+(v) = 0$ for $v > \rho$; $s^- = 0$ if b is even, and $s^-(v) = 0$ for $v \ge \rho$ if b is odd; $S_2(s) \le (1 + \epsilon_b)N$; and, if b is odd, $\sum_v (s^+(v) - s^-(v))(v^L - v)$ is 0 for odd L with $3 \le L < b$ and $2b$ for $L = b$ (compare Lemma 2.4 and (2)). With the *cycle filter*, also $\sum_v s^+(v)$ is at least the number $2\lceil b/6 \rceil - 1$ of eigenvalues of C_b greater than 1, and for odd b, $\sum_v s^-(v)$ is at least the number of eigenvalues of C_b less than -1 (compare Lemma 4.1).

A sequence passes the *moment test* if some feature vector s, admissible for (b, N), has $M_4(s) = \Theta(G) + 4[b = 4]$ and $S_2(s) \le n$ if b is even, and $M_4(s) = 2\Theta(G)$ and $S_2(s) \le 2n$ if b is odd; with the *load bound*, also $n - \max(Q, 2) \le S_2(s)$ (for odd b, $2n - 2\max(Q, 2) \le S_2(s)$), compare (1) and Lemma 2.4(c).

$\operatorname{Cand}_B(b, N)$ is the set of sequences for (b, N) that pass the moment test with the load bound and without the cycle filter (the formal model of the moment test of B, version 1, and, for even b, of D, version 2). Its subset $\operatorname{Can}_B(b, N)$ consists of the sequences that are lexicographically least among their images under the dihedral group of $\mathbb{Z}_b$, comparing the keys $1000p_k + q_k$ for $k = 0, 1, \dots$ in turn, and $\operatorname{Can}_D(b, N)$ of those that are greatest, for the keys $64(p_k + 2q_k) + p_k$.

$\operatorname{Cand}_{B3}(b, N)$ (the formal model of version 3 of B) is the set of sequences that pass the moment test with the cycle filter, the load bound and, for even b, the inequalities $0 \le 2S_2(s) - n - 2\sum_v s^+(v) \le \max(P, 2)$ of Lemma 4.2; whose key $1000p_0 + q_0$ is at least that of every other vertex; and that pass, for every d with $1 \le d \le b$, the window of Lemma 5.3 with some feature vector s admissible with the cycle filter (for $d = b$ the upper end of the window is Θ_d itself; for even b, s must also satisfy $\max(b + P_d + 2Q_d, S_2(s)) \le n_{\max}$ and the inequality of Lemma 5.4, with $S_2(s)$ and $\sum_v s^+(v)$ in place of S_2 and s). The leaf set of C at (b, N), with the rank bound, consists of the sequences with $\sigma(r_0 - 4[K_q = \emptyset]) \le 2N$, in the notation of Lemma 4.4, that are least among their dihedral images for the keys $[p_k > 0] + 2[q_k > 0]$.

For each modelled version of each program (A, versions 4 to 6; B, versions 1 to 3; C and C2; D, versions 1 to 3), the statement files define in the same way the leaf set at (b, N), from the node test of A (Lemma 5.1 with the bounds described after it, Lemma 2.4(c) and (1), and optionally Lemma 5.4), the window of Lemma 5.3, the rotation rules, canonical forms and moment tests of B and D, and the rules of C. D's pruning is modelled by recursively defined tables $\operatorname{reach}_1(r, w, t)$ and $\operatorname{reach}_2(r, w, Q, t)$ for $t < 3072$ and a predicate Reach_D on partial assignments, which is true if some multiset of eigenvalues of the required shape is compatible with some completion.

Proposition 5.7. *(a) Every admissible feature vector is dominated, in* M_4*,* S_2 *and the numbers of values greater than* m *and at least* m *for* $1 \le m \le 9$*, by one that is maximal for this order. (For odd* b *these numbers are the smaller of the counts on the positive and on the negative side, and the number of values at least* m *is taken to be* 0 *for* $m = 1$*.)*

(b) *For $b \le N$, the recursive enumeration of the statement files produces exactly the sequences for (b, N).*

(c) *Let $3 \le b \le N \le 49$ and let $G = G(p, q)$ be integral, with (p, q) a sequence for (b, N). Then $s^+(v) = \mathrm{mult}(v)$ and, for odd b, $s^-(v) = \mathrm{mult}(-v)$ (for even b, $s^- = 0$) define a feature vector that satisfies all conditions of admissibility except possibly the cycle filter.*

(d) *For $N \le 49$, the node test of A is monotone: if it holds at depth $d < b$ for (p', q') and $p \le p'$, $q \le q'$ componentwise, it holds for (p, q).*

(e) *Let $3 \le b$ and $N \le 49$. If (p, q) is a sequence for (b, N) and G is integral, then (p, q) passes the window test with the cycle filter at every depth $d \le b$.*

(f) *For $t < 3072$, $\mathrm{reach}_1(r, w, t)$ holds if and only if some pair of maps $\mathbb{Z}_r \to \mathbb{N}$ has weight $P + 2Q = w$ and Θ-sum t, and likewise reach_2 with Q prescribed; and, for even b and $b \le N \le 49$, Reach_D at a partial assignment of $v_0, \dots, v_{d-1}$ whose assigned weight is at most $N - b$ holds if and only if some completion with $n \le N$ passes the moment test.*

(g) *For every modelled version and every (b, N) in its domain ($3 \le b \le N \le 49$, with bounds on b for each version), every integral G with (p, q) a sequence for (b, N) has a dihedral image in the leaf set of that version.*

(h) *For every modelled version and (b, N) in its domain, a sequence in the leaf set has a factorisation of $\det(xI - A)$ into factors $x - z$, $z \in \mathbb{Z}$, if and only if G is integral; and, for a sequence (p, q) for (b, N), G is integral if and only if some dihedral image of (p, q) is such a leaf.*

Proposition 5.8. *(a) For odd b with $5 \le b \le 15$ and $N \in \{41, 42\}$, no feature vector is admissible for (b, N).*

(b) *For $b \in \{22, 26\}$ and $N = 42$, no feature vector is admissible with the cycle filter, and Reach_D (with the load bound, without the cycle filter) is false for the empty assignment; so, by Proposition 5.7(f), no sequence with $n \le 42$ passes the moment test with the load bound.*

(c) *$\mathrm{Cand}_{B3}(14, 42)$ and $\mathrm{Cand}_{B3}(18, 42)$ are empty, and so is the leaf set of C with the rank bound at $(13, 41)$.*

Proposition 5.9. *(a) $\mathrm{Cand}_B(4, 41)$ has 410,544 elements, and $\mathrm{Can}_B(4, 41)$ and $\mathrm{Can}_D(4, 41)$ have 52,641 each. $\mathrm{Cand}_B(6, 42)$ has 14,215,675 elements, and $\mathrm{Can}_B(6, 42)$ and $\mathrm{Can}_D(6, 42)$ have 1,187,991 each.*

(b) *There are 3,031,469 sequences for $(3, 49)$ and 1,246,783 for $(3, 42)$.*

Proposition 5.9(b) counts all labelled sequences, without symmetry reduction or pruning. In version 1 the numbers of Proposition 5.9 were reported as numbers of sequences passing the moment tests of B and D and tested by them; here they are only the sizes of the formal sets.

6 Small cycle lengths and the proofs of Theorems 1 and 2

The following three classifications were obtained in version 1 only as results of the computer searches. We have no written proofs of them; they are proved in Lean (Section 8, where the routes of the formal proofs are outlined).

Proposition 6.1. *Every integral generalized sun graph with cycle length 3 that is not a cycle has at least 50 vertices.*

Proposition 6.2. *The images of* $(0,6,6,12,6,6)$ *under the dihedral group of* $\mathbb{Z}_6$ *are its six rotations, which are distinct. Let* $b=6$ *and let* $p,q\colon \mathbb{Z}_6\to\mathbb{N}$*, not both zero, with* $n\le 42$*. Then* $G(p,q)$ *is integral if and only if* $q=0$ *and* p *is one of these six rotations.*

Proposition 6.3. *An integral generalized sun graph with cycle length* 4 *that is not a cycle and has at most* 49 *vertices is isomorphic to* $C_{4,1}(6,0,3,0)$*,* $C_4(5P_2,P_1,2P_2,P_1)$*,* $C_{4,2}(4,4,0,0)$ *or* $C_{4,2}(10,10,0,0)$*, and these four graphs are integral. The last one has* 44 *vertices, characteristic polynomial* $x^2(x^2-1)^{19}(x^2-9)(x^2-16)$ *and spectrum* $\{0^{[2]},\pm1^{[19]},\pm3,\pm4\}$*.*

Let H be a generalized sun graph with cycle length b whose pendant paths have at most two vertices. Reordering the paths at each cycle vertex shows that H is isomorphic to a graph $G(p,q)$ of Section 2, and H is connected, with n vertices and n edges, so it is unicyclic and its cycle has length b.

Proof of Theorem 1. Let G be a counterexample with $n\le 42$; by [2, Theorem 2.1] it carries only copies of P_1 and P_2, so we may assume $G=G(p,q)$. If $b=3$, then $n\ge 50$ by Proposition 6.1. Odd $b\ge 5$ is excluded by Corollary 3.1, as it would give $n\ge 102$. If $b\equiv 2 \pmod 4$, then $b\in\{6,10,14,18\}$ by Lemma 4.1, and $b\notin\{14,18\}$ by Corollary 4.3. If $b=6$, Proposition 6.2 gives $q=0$ and a rotation p of $(0,6,6,12,6,6)$; the rotation of the indices is an isomorphism $G\cong C_{6,1}(0,6,6,12,6,6)$. This graph is a counterexample: it is integral by [2, Proposition 3.2], has 42 vertices, and $4\nmid 6$. The remaining case $b=10$ is excluded only in the formal proof of Theorem 1; we give no written proof of it. As isomorphic graphs have the same number of vertices, every counterexample has at least 42 vertices. □

Proof of Theorem 2. Let G be integral, not a cycle, with $n\le 41$ and cycle length b. If $4\nmid b$, then G is a counterexample with $n\le 42$, so by Theorem 1 it is isomorphic to $C_{6,1}(0,6,6,12,6,6)$, which has 42 vertices; this is impossible. So $4\mid b$, and by [2, Theorem 2.1] and Lemma 4.1, $b\le 12$. If $b=4$, Proposition 6.3 gives one of four graphs, and the one with 44 vertices is excluded. The remaining cases $b=8$ and $b=12$ are excluded only in the formal proof of Theorem 2; we give no written proof of them. □

7 Open questions

The questions of [2, Section 5], whether the cycle length b of an integral generalized sun graph other than a cycle must be even, and whether 4 and 6 are the only possible values, are not answered here. By Corollary 3.1, an odd b is 3, or at least 15 with $n\ge 102$; and $b=3$ needs $n\ge 50$ (Proposition 6.1). Lemma 4.4 bounds b for either parity: if $r_{\max}(n)$ is the largest t with $\sigma(t)\le 2n$, then $b\le r_{\max}(n)+4$; writing $r_{\max}(n)=4k+j$ with $0\le j\le 3$, we have $2n\ge\sigma(4k)\ge 4\sum_{v=1}^{k}v^2\ge\frac{4}{3}k^3$, so $b\le(96n)^{1/3}+7$.

Thanks are due to Rodrigo O. Braga, Jean Carlo Moraes and Matheus C. Santos for the question answered here.

8 Formal verification

All the results are formally verified in Lean 4 (version 4.34.1) with Mathlib [9] (commit `d13f23b7`). Version 1 of this paper had no formal verification. For version 2 every result of the paper was formalised: each of its mathematical claims is a statement of Table 1, a definition, a step of a written proof of one of them, or an observation that is immediate from the definitions. Where the paper gives no written proof (Propositions 5.5–6.3, the case $b=10$ of Theorem 1 and the cases $b\in\{8,12\}$ of Theorem 2), the result rests on the formal proof alone.

Statements before formal proofs. The statements were written from version 1 of the paper and reviewed independently before the formal proofs were accepted. On 27 September 2026 a Claude Code agent wrote a first Lean statement file for version 1, with 25 statements and 73 known-answer tests of its definitions, and an independent Claude Code agent reviewed it. OpenAI Codex then wrote the present statement files, which keep those 25 statements word for word and add 35, among them the statements on the search rules of version 1 in Section 5; 120 named known-answer tests check their definitions. A Codex agent that had not written the files reviewed them, the files were compiled in two new empty directories, and they were frozen on 8 October 2026 at 10:43 UTC. On 9 October 2026 a Lean proof showed that one of the 60 statements, `LegacyTransferCRTExactness` (Proposition 5.6), was false as frozen: in the definition of the bound E a missing pair of parentheses made it $\prod_k |S_k| + |h|$ instead of $\prod_k (|S_k| + |h|)$, and $b = 3$, $p = (4, 4, 4)$, $q = 0$, $m = 2$ is a counterexample. On 18 October 2026 the author approved adding exactly this pair of parentheses; the corrected files, identical to the frozen ones apart from these two characters and with seven added regression tests, were reviewed and validated in the same way and frozen again on 18 October 2026 at 22:28 UTC. The earlier files and the counterexample are kept.

Two of the 60 statements, `LegacyDump52652` and `LegacyA4DumpCompleteness`, are not part of version 2. They concern two fixed files of 52,652 sequences each, written at $b = 10$, $N = 41$ by version 4 of A and version 1 of B2, and assert that the two files are equal as sets, that one of them is the formal leaf set of A at $(10, 41)$, and that none of the 52,652 graphs is integral. Their proofs were not completed (in five attempts the Lean kernel could not read in the two data files, of about 2.5 MB each), and on 3 November 2026 the author decided to delete them, together with every claim of version 1 that only they would support. They remain in the frozen files, unproved; nothing in this paper depends on them or claims them.

Formal proofs. The proofs are in separate files that import the statement files and do not modify them. From 8 to 18 October 2026 OpenAI Codex proved 57 of the statements, in 31 groups; each group was accepted after builds in two new empty directories and a review of the evidence by a Codex agent that had not written it. For the remaining statement, `LegacyMomentCounts` (Proposition 5.9(a)), Codex proved the numbers at $(b, N) = (4, 41)$ in October 2026, and by 3 November 2026 it had written the reduction of the numbers at $(6, 42)$ to the size of $\mathrm{Cand}_B(6, 42)$, and the sizes of eleven of the 21 fibres of that set described below and part of a twelfth. On 3 and 4 November 2026 Claude Code (model Claude Opus 5.5) wrote the remaining 190 slice modules and 10 fibre modules, the module that assembles `LegacyMomentCounts`, and the final package.

What is proved. The final package (module `ClaudePaper7FinalPackage`, namespace `Paper7FinalPackage`) restates each of the 58 statements of Table 1 with its exact frozen type, as a theorem `final_X` of type `X`, proved by the theorem of the proof files that proves `X`, and prints its axioms.

Routes of the formal proofs. We outline the formal proofs only where the paper gives no written proof; elsewhere the formal proofs may follow other routes than the written ones. *Theorem 1.* The formal proof reduces a counterexample with $n \le 42$ to the cycle lengths 6 and 10, settles $b = 6$ by Proposition 6.2, and excludes $b = 10$, after showing by Lemma 4.4 that no pendant path P_2 occurs, by a case analysis of the multiplicities of the eigenvalues $2, \dots, 6$ allowed by the lemmas of Sections 2 and 4, combined with congruences modulo 4 obtained from the transfer matrices of Lemma 2.2(a). *Theorem 2.* The formal proof excludes $b = 8$ and $b = 12$ by case analyses of the possible spectra, using transfer-matrix identities, rank bounds and parity arguments on the supports of p and q. *Proposition 6.1.* The formal proof reduces the case $b = 3$ to three residual configurations, and for each shows, by a kernel-checked

search over the possible integer root patterns of a polynomial attached to the graph (a table of candidate rows whose completeness is itself proved in Lean), that every pattern is rejected by an explicit identity. *Propositions 6.2 and 6.3.* After the reduction to pendant paths with at most two vertices, the formal proofs exclude or identify the graphs by case analyses of the possible spectra (for Proposition 6.2, the largest eigenvalue $\rho \in \{3, 4, 5, 6\}$) and of the positions of the pendant vertices and paths; the integrality of the graphs found is proved from their characteristic polynomials. *The classification of unicyclic graphs with at most* 21 *vertices* (Section 1) is proved for arbitrary unicyclic graphs: the formal proof shows that in an integral one with at most 21 vertices every vertex off the cycle has degree at most 2, so that it is a generalized sun graph, and then uses the results on generalized sun graphs. *Proposition 5.9(a).* At $(6, 42)$, Burnside's lemma for the dihedral group of order 12 reduces the sizes of Can_B and Can_D to the size of $\mathrm{Cand}_B(6, 42)$ and to the numbers of sequences fixed by the reflections and rotations, which are proved separately. The set $\mathrm{Cand}_B(6, 42)$ is split into 21 fibres by the value of $\Theta(G)$; the ten fibres completed by Claude Code were split into slices by the value of Q, and the size of each slice was computed by closed reduction in Lean's metaprogramming layer, and the value is checked by `rfl` in the kernel.

Checks. The Lean package of version 2 has 1925 modules: the frozen statement and test files, the proofs and the final package. It was built module by module from an empty directory on the author's Windows computer, and every module was checked with `leanchecker` (4 November 2026, 04:09–11:39 UTC); all 3850 compile and `leanchecker` commands ended with exit code 0, with no error, no warning and no `sorry`. In a first pass, run with 13 parallel jobs, seven modules failed within the same second after about 13 seconds, and the 25 modules that depend on them were skipped; for one of the seven, the log read at the time showed that Lean could not read the compiled file of a dependency that had been built three hours earlier. The 32 modules were then rebuilt from the same sources with 8 parallel jobs, and all passed; the logs of the failed attempt were overwritten by the rebuild. All 85,707 axiom lists printed by `#print axioms` are subsets of `propext`, `Classical.choice` and `Quot.sound`. A static scan of the sources, outside comments, found no `sorry`, `admit`, `native_decide`, `bv_decide`, `implemented_by`, `extern` or `unsafe`, and no `axiom` declaration. A second build from another empty directory followed on the same computer, with the same sources (4 November 2026, compilation 11:39–16:52 UTC, `leanchecker` 16:52–22:08 UTC). All 1925 modules compiled with no error, no warning and no `sorry`. The `leanchecker` pass did not run in one piece. Its driver script stopped at 17:31 UTC after 425 modules, for a reason we did not establish, and was resumed at 19:42 UTC from the modules not yet checked. Then, at about 21:41 UTC, ten modules failed within four seconds because Lean could not read the compiled file of a dependency, and the 25 modules that depend on them were skipped. No module was rejected by the kernel. The Windows system log reports low virtual memory in the same seconds, and a later run with ten parallel jobs was seen to use up almost all of the memory that Windows can commit, so we attribute these failures to the 12 parallel jobs, although the memory was not measured at the time. The 35 modules were checked again with 6 parallel jobs, from the same compiled files, and all passed; the logs of the failed attempts are kept. This build also gives 85,707 axiom lists, all subsets of the same three axioms, and its 1925 compiled `.olean` files are byte for byte identical to those of the first build. The whole project, with the other papers of the repository, was then rebuilt from a third empty directory, with 260 warning lines in older modules, most of them deprecations (22:08 UTC on 4 November to 08:57 UTC on 5 November 2026). All 2841 modules compiled and 2840 passed `leanchecker`. The remaining module consists of `import` lines and has no constants, so there is nothing to replay in it; `leanchecker` on it was not completed. The first attempt ended when the computer crashed, after about an hour and a half, for a reason we did not establish, and we stopped the second at 69 GB of memory. Seven modules ran out of memory during this rebuild, the 16 modules that depend on them were skipped, and these 23 were built again and passed.

No Lean check rejected a module in any of the three builds.

Lean bounds the work of each elaboration step by a number of "heartbeats". In the 190 slice modules and the module that assembles `LegacyMomentCounts`, written by Claude Code, this limit is removed (`set_option maxHeartbeats 0`) for the computation of the counts and their checks in the kernel (the values were found by reduction in the metaprogramming layer and are only trusted after the kernel has re-derived them by `rfl`). It is raised to 800,000 in eight proof modules written by Codex (prime certificates used for Propositions 5.5(f) and 5.6(a)), and set explicitly, to 200,000 or 1,000,000, in three test files of the frozen statements. The limit only bounds how long Lean may work on a step; it does not change what is accepted. The soundness of the proofs rests on the Lean kernel, which checks every proof term, as `leanchecker` did again for every module.

The statement files, the tests, the proofs and the final package are in version 1.14.0 of the repository [6], directory `lean/Research/Backfill/Paper7/` (the copies of the frozen files differ from them only in their `import` lines), and the receipts of the builds in `certificates/sun-graphs/lean-package-v2/`.

Definitions the reader must trust. From Mathlib the statements use finite simple graphs, their adjacency matrices over $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$, graph isomorphisms, walks and cycles, the cycle graph C_b, characteristic polynomials, real roots of polynomials with multiplicity, determinants, traces, ranks and kernels of matrices over $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{Z}/\ell$, integer polynomials and their reductions modulo a prime, primes, Bernoulli numbers, finite sets and cardinalities. The statement files define: integral graphs (all complex eigenvalues are integers), the spectrum as the multiset of real roots of the characteristic polynomial, mult, $N_{>t}$, $N_{\geq t}$, $N_{<t}$, $N_{\leq t}$, n_+, n_-, n_0, B^+, S_2, numbers of cycles of a given length, the odd girth, unicyclic graphs with a cycle of length b, κ_b and $\sigma(t)$; generalized sun graphs $C_b(L)$ for an arbitrary list $L(k)$ of path orders at each vertex, their order, their number of pendant vertices and counterexamples; the graphs $G(p,q)$, P, Q, K_p, K_q, Θ, f_k, $S(\lambda)$, R, R', T_k, M, gaps, δ_0, $\delta_{\pm 1}$, X, tridiagonal matrices, the pivot count of Lemma 5.2 and $g(w)$; the named graphs of the paper and the two families of Section 1; the polynomial F, the systems and offsets of Section 5.2, the interpolation nodes, divided differences and Newton reconstruction, the chain recurrence of Proposition 5.5(h), the primes of Propositions 5.5 and 5.6, the integer transfer matrices, E and the residue tests of A; and the formal search rules of Section 5.3: feature vectors, admissibility, the moment tests, the window, the node test of A, rotation rules, canonical forms, the rules of C, the tables of D, the enumeration, the domains and leaf sets of all modelled versions, the list of the 23 runs of A of Proposition 5.6(c), and the sets of Propositions 5.8 and 5.9. The 120 known-answer tests check these definitions on examples.

Faithfulness and non-claims. In the formal statements a generalized sun graph is given by b and, for each cycle vertex, a list of the numbers of vertices of the paths attached to it; the results stated for arbitrary generalized sun graphs (Theorems 1 and 2, [2, Theorem 2.1], Corollary 4.3 and Propositions 6.1 and 6.3) are formalised in this generality, and the others for the graphs $G(p,q)$. Corollary 4.3 is formalised also for the cycles C_{14} and C_{18}. The statements of Section 5 concern formally defined sets and predicates modelled on the rules of the programs of version 1; no statement says that the programs implement them, that their runs were complete, or what they printed. B2, which used floating-point arithmetic, is not modelled. Passing a test of Proposition 5.5(f) is not claimed to imply integrality. The 23 runs of Proposition 5.6(c) are a list of data taken from version 1. The identifications of the families of Section 1 with those of [2] and [13], and of the graphs of Theorem 2 with those of [4], are bibliographic and not formalised; so are Conjecture 2 as a conjecture of [1] and the questions of Section 7. That a generalized sun graph whose paths have at most two vertices is unicyclic and isomorphic to some $G(p,q)$, used in the written proofs of Section 6, is not a separate formal statement; the formal proofs of

the results that use it prove what they need. The agreement of the Lean definitions with those of the paper was checked by the reviewing agents, not by a machine.

Changes in version 2. Apart from this section, Section 5, Section 6, the abstract, the disclosure and the dates, version 2 differs from version 1 as follows. The method paragraph of Section 1 no longer says that every case is confirmed twice by hand proofs or programs, and the paragraph on prior work is shortened. Section 1 now says that the classification of [4] up to 21 vertices is verified in Lean, and adds that the three graphs of Theorem 2 are pairwise non-isomorphic, that the graph named by Veras is integral, the second interval of [2, Theorem 2.1], and the restriction $a \geq 1$ for both families (for $a = 0$ they are the cycles C_6 and C_4); Theorem 2 is now said to exclude other integral generalized sun graphs up to 41 vertices that are not cycles. The written proof of Corollary 4.3 now treats the cycles C_{14} and C_{18} and the reduction to pendant paths with at most two vertices. In Remark 3.2 the word "only" before the classical congruence and the numbers of searches are dropped. In Section 5, the descriptions of the programs, Table 1 (the searches), Table 2 (program versions and hashes), the controls and cross-checks, and the statement that the programs decide integrality exactly are deleted; the cells of Table 1 that said "no spectrum" or "0" and five of its numbers (and one number of its Section 5.3) are restated as Propositions 5.8 and 5.9 about the formal sets; the remark that no overflow occurred and the remark that the window is not monotone are deleted; D's bounds at 0 and ± 1 now include the offsets $P - |K_p|$ and $Q - |K_q|$, which version 1 omitted; and the statements about spectral windows and sieves now carry the hypothesis $n \leq 49$. Deleted for lack of a formal proof: the statement that versions 4 of A and 1 of B2 produce the same set of 52,652 sequences at $b = 10$, $N = 41$, none of them integral, and the numbers 52,652 and 14,662 of Table 1; the agreement of the numbers of B2 and A at $b = 6$, $N = 42$; the counts of search-tree nodes, leaf tests and tested sequences of Table 1 and of Section 5.3 of version 1; the random tests on 2,968, 400, 3,877 and 3,250 graphs; the run of D at $b = 30$ in Table 3; and the run times. Table 3 (confirmations of each case) is replaced by the written proofs of Section 6, which now cite Propositions 6.1–6.3 and Theorem 1 instead of the searches; Lemma 5.4 and Propositions 5.5, 5.6 and 5.7 state explicitly what version 1 used informally. Section 7 cites Proposition 6.1 instead of the searches, and no longer says that the questions of [2] remain open. Apart from the deletions and additions listed here, the statements of the remaining results are unchanged.

Tool and computational resource disclosure

Version 1 of this paper was produced on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. Claude Code agents selected the problem, found the results and their proofs, wrote the programs A and B, carried out the literature searches described in the introduction, and drafted the text. Two independent Claude Code agents, which had not produced the results, refereed version 1; each wrote a search program of its own (C and D), and their corrections were applied. Version 1 rested on the written proofs and on these computations; it had no formal verification.

Version 2 was produced from 27 September to 5 November 2026. On 27 September 2026 a Claude Code agent (model Claude Opus 5.5) wrote the first statement file of Section 8, and an independent Claude Code agent (model Claude Opus 5.5) reviewed it. OpenAI Codex (desktop application; its main session used the model `gpt-6-astra`) wrote the present statement files, which a Codex agent that had not written them reviewed before they were frozen on 8 October 2026, and the corrected files frozen on 18 October 2026; from 8 to 18 October 2026 Codex wrote the Lean proofs of 57 statements, and until 3 November 2026 part of the proof of the remaining statement, `LegacyMomentCounts`. Each group of proofs was accepted after two builds from empty directories and a review of the evidence by a Codex agent that had not written it. On

3 and 4 November 2026 Claude Code (model Claude Opus 5.5) wrote the rest of the proof of `LegacyMomentCounts` and the final package, which was built and checked on the author's Windows computer (Section 8). Claude Code (model Claude Opus 5.5) compared version 1 claim by claim with the statement files and drafted the text of version 2. An independent Claude Code agent (model Claude Fable 5.1), which had written none of the files, reviewed the draft against the statement files and the evidence of the first build on 4 November 2026. On 5 November 2026 a second independent agent (same model) compared the final package with the frozen statements, read the proof of `LegacyMomentCounts` for steps that avoid the kernel, recounted the evidence of the three builds and reviewed the paper; neither ran Lean. The corrections of both reviews were applied. Throughout, "we" refers to this workflow; the reading and searching described in the paper were done by the agents. The author is not a professional mathematician. The author has read version 2 of this paper, but did not check its proofs and did not compare its Lean statements with the paper. No human mathematician has reviewed this paper. The correctness of the results rests on the Lean proofs of the statements in Table 1 and on the agreement of those statements with the paper, which was checked by the reviewing agents; the written proofs are given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

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In the paper	Lean statement
Theorem 1	`Theorem1`, `Theorem1MinOrder`
Theorem 2; orders and spectra of its three graphs, and that they are pairwise non-isomorphic (Section 1)	`Theorem2`, `Theorem2Graphs`, `Theorem2GraphsDistinct`
Proposition 3	`Proposition3`
Corollary 3.1, Remark 3.2	`Corollary31`, `Remark32`, `TracePowerPrime`
[2, Theorem 2.1] and its consequence for generalized sun graphs	`BMSTheorem21General`, `BMSTheorem21`, `OnlyP1P2`
[2, Proposition 3.2]: the spectrum of $C_{6,1}(0,6,6,12,6,6)$	`BMSProposition32`
Section 1: unicyclic graphs with at most 21 vertices	`IntroUnicyclic21`
Section 1: infinite families; $C_{6,1}(0,a,a,2a,a,a)$; $C_{4,1}(0,a,2a,a)$	`IntroInfiniteFamilies`, `IntroHexagonInfiniteFamily`, `IntroVeras68`
Section 2: n, edges, $\Theta(G)$, isomorphism classes	`SetupFacts`
Lemmas 2.1–2.4, (1) and (2)	`Lemma21`, `Lemma22`, `Lemma23`, `Lemma24`, `Identities12`
Lemmas 4.1, 4.2, Corollary 4.3, Lemma 4.4	`Lemma41`, `Lemma42`, `Corollary43`, `Lemma44`
the constants in the proofs of Lemma 2.4, Corollary 3.1 and Lemma 4.4, and $\sigma(9), \sigma(10), \sigma(11), \sigma(14)$	`ArithmeticConstants`
Lemmas 5.1–5.4	`Lemma51`, `Lemma52`, `Lemma53`, `PartialAssignmentCounting`
Proposition 5.5(a), (b)	`SpectrumMultiplicityBridge`, `SunSpectralWindowAndFactorization`, `IntegralMultiplicitySum`
Proposition 5.5(c), (e), (f), (g)	`ClearedCharacteristicPolynomial`, `ModularPolynomialRejectSound`, `LegacyModuliAndWindows`, `LegacyPolynomialSieveSoundness`, `ModularInterpolationExactness`
Proposition 5.5(d), (f)	`ReducedMultiplicityExactness`, `RankReductionAndNullity`, `ModularNullityUpperBounds`, `LegacyNullitySieveSoundness`
Proposition 5.5(h)	`UnicyclicSchwenkIdentity`, `SunSchwenkIdentity`, `SchwenkRecurrenceExactness`
Proposition 5.6	`LegacyTransferCRTExactness`
Proposition 5.7	`PruningAndEnumerationCompleteness`
Proposition 5.8	`LegacySpectrumEmptyCells`, `LegacyPruningEmptyCells`
Proposition 5.9	`LegacyMomentCounts`, `LabeledBudgetCounts`
Propositions 6.1, 6.2, 6.3	`Section6CycleThree`, `C61OrbitSix`, `ControlB4N49`, `ControlC42Ten`
the bound of Section 7	`Section6Bound`

Table 1: The 58 statements of version 2. The statements of Sections 1–4, Lemmas 5.1–5.4, Section 6 and Section 7 are in the namespace `BackfillPaper7.Challenge`; those of Propositions 5.6–5.9(a) in `BackfillPaper7.Challenge.Legacy`, `LabeledBudgetCounts` (Proposition 5.9(b)) in `BackfillPaper7.Challenge`, and those of Proposition 5.5 in `BackfillPaper7.Challenge.Legacy.Algebra`. The names `Section6Bound` and `Section6CycleThree` refer to the section numbering of an early draft.